Elevating LLMs via synergy of Quantum Computing and classical transformer models

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2025 5th International Conference on AI-ML-Systems (AIMLSystems) | 979-8-3315-7318-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/AIMLSYSTEMS67835.2025.11330851

***Abstract*—Quantum computation offers new directions in computer science that go beyond classical systems' state-of-the-art abilities. The combination of quantum technologies with Large Language Models (LLMs) provides promising avenues for the creation of more efficient and powerful NLP architectures. In this research, we examine quantum-enhanced transformers and further develop the Self-Attention Sequential Quantum Transformer Channel (SASQuatch) framework to enhance inference time and perplexity. The models proposed are tested on sentence completion tasks and compared against a baseline transformer, showing significant gains in efficiency and predictive power. We further compare the method with other optimization methods, such as the Quantum Approximate Optimization Algorithm (QAOA), thus highlighting the necessity of further investigation of hybrid quantum architectures in improving the performance of large-scale language models.**

***Index Terms*—Quantum Self-Attention, Quantum Machine Learning, Quantum Artificial Intelligence, Quantum Computing, Quantum Approximate Optimisation Algorithm**

I. Introduction

The rapid development of Generative Pre-trained Transformers (GPT) [1] and Large Language Models (LLMs) [2] [3] has revolutionised natural language processing (NLP). However, these advantages come at the price of ever-rising computing demands that are beginning to approach the physical and energy limitations of conventional technology. By enabling speedups and more efficient optimisation methods, quantum computing has demonstrated promise in overcoming these challenges through the use of ideas like superposition and entanglement.Here, we investigate the Self-Attention Sequential Quantum Transformer Channel (SASQuaTCh) [4], a hybrid quantum-classical architecture designed to reduce the computational and temporal complexity of LLMs while maintaining language quality. By optimising linear layers using the Quantum Approximate Optimisation Algorithm (QAOA) [5] and incorporating quantum embeddings via Variational Quantum Circuits (VQCs) [6] within the transformer's self-attention mechanism, SASQuaTCh offers a practical method of lowering classical overhead without sacrificing performance. We demonstrate its efficacy in autoregressive text creation with notable improvements in perplexity [7] and inference time across multiple LLM scales.

Recent developments in quantum machine learning (QML) [8], particularly in parametrized circuits and quantum kernel approaches in high-dimensional Hilbert spaces, are expanded upon by this technique. Due to several significant advances, the quantum-enhanced architecture is more efficient. These include quantum layers with lower parameter complexity through compact quantum state encoding of attention weights and faster inference using QAOA based linear transformations. Beyond improvements in performance, the potential influence might shift the course of generative AI by opening the door to more energy-efficient language models.

Experimental studies show that quantum-enhanced LLMs outperform traditional benchmarks and provide computing advantages. Extensive testing across small, medium, and large GPT-2 variants shows measurable improvements in processing speed and model correctness. These findings validate the viability of quantum-enhanced NLP systems and show significant promise for scaling quantum-language models. These hybrid approaches have the potential to revolutionize large-scale language modelling as quantum hardware continues to progress by removing conventional processing barriers and opening up new avenues for AI efficiency and capabilities. Our work focuses on not only improvising the time inference of generation but also at the same time improvising the perplexity of the models thereby, a well timed output with lesser hallucinations in the model.

Additional refinement of these LLMs through Quantum computing would make the performance of these hybrid models even better, outperforming existing output levels. Different optimizations of LLMs through quantum computing, such as the Ansatz Architecture Search architecture, which is primarily derived from cutting-edge research of VQAs, also contribute to further improvement.

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II. Related Works

The integration of quantum computing with transformer-based language models has been possible with the help of various advances in the field of Quantum Machine Learning (QML) and hybrid architectures. Developments in QNLP (Quantum Natural Language Processing) [9], exploring quantum embeddings and variational quantum circuits for text classification, demonstrate more efficient results by quantum kernels than classical methods. The Self attention Sequential Quantum Transformer Channel (SASQuaTCh) [4] introduces a variational quantum circuit design which helps in implementing self attention mechanism. Our approach is an extension of this project wherein we tried to optimise it further to obtain a better perplexity on top of a better time inference. The method introduced quantum enhanced token embedding and Quantum Fourier Transform [10] which resulted in exponential improvements in the runtime complexity as well as runtime complexity. Quantum Approximate Optimization Algorithm (QAOA), one of the adapted methods, is applied for tasks like graph-based optimization and neural network compression. QAOA is used to optimize the attention weights that avoids relying on costly gradient updates. Haque et. al [11] presents a machine learning approach to improve the efficiency of Chronic Kidney Disease (CKD) detection by leveraging a fine-tuned CatBoost model enhanced with nature-inspired algorithms and Explainable AI (XAI). Among the models evaluated—Logistic Regression, Multi-Layer Perceptron, Random Forest, and CatBoost—the fine-tuned CatBoost model achieved the highest performance, with a test accuracy of 98.75%, an AUC of 0.9993, and a Kappa score of 97.35%.

Quantum circuits can replace classical submodules, without sacrificing accuracy, as seen from prior hybrid architectures like the Quantum Convolutional Neural Networks (QCNNs). A similar method is considered in the SASQuaTCh approach, which focuses on transformer-specific optimizations instead of circuits. SASQuatch extends these methods like amplitude embedding and residual quantum connections for efficient quantum state initialization and stabilize training, respectively. Low-Rank Adaptation (LoRA) further reduces transformer memory usage, which is put to use by SASQuaTCh in various ways, like replacing dense linear layers with QAOA-optimized circuits, dynamic quantum-classical switching like an efficient hybrid model, in order to achieve better performance without any constraints and limitation in hardware.

LLMs are used to generate quantum circuits in recent works, as can be seen for QAOA and VQE, and SASQuaTCh aligns with this trend via automation of quantum circuit integration in classical architectures with the help of PyTorch and Qiskit, along with parameterization of circuits based on hidden states of transformers.

III. Methodology

In this study, we present a hybrid quantum-classical strategy that integrates quantum computing notions into a classical transformer framework to enhance large language models (LLMs). The SASQuaTCh quantum-enhanced transformer architecture achieves this by combining optimised quantum-classical inference that uses dynamic parameterised quantum circuits (PQCs) to improve logit generation while maintaining computational efficiency, hybrid attention mechanisms that enhance contextual modelling by combining variational quantum circuits and classical self-attention, and quantum embedding layers that use high-dimensional Hilbert spaces to provide richer semantic representations. By enabling the model to leverage quantum parallelism and entanglement for improved linguistic feature extraction while operating within traditional computational restrictions, these enhancements bridge the gap between theoretical quantum benefits and practical NLP applications.

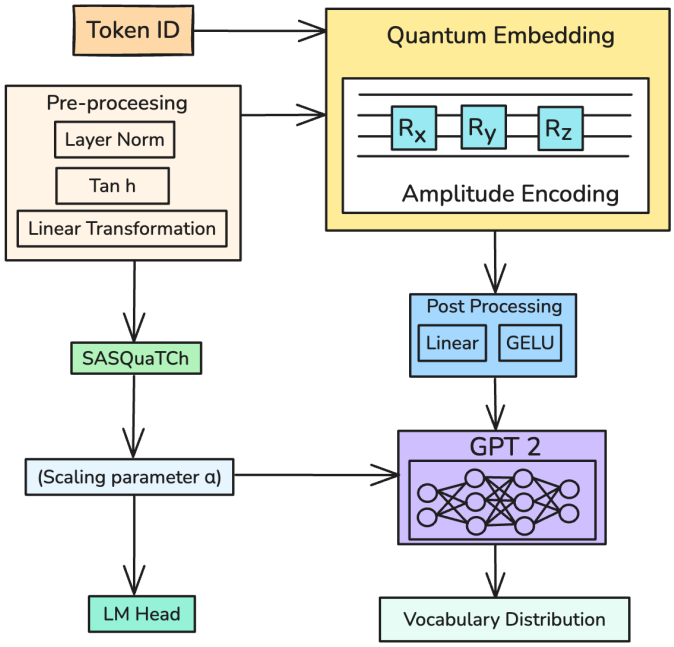


Fig. 1. Architecture of proposed model

A. Model Architecture

1) SASQUATCH Model: In this paper, We evaluate a hybrid quantum-classical approach to improve the performance of large language models (LLMs) by integrating concepts from quantum computing into a classical transformer architecture. Figure 1 illustrates the integration of a quantum-enhanced embedding pipeline with a frozen GPT-2 model. Once the input prompt has been tokenised using the standard GPT-2 tokeniser, a preprocessing pipeline normalises and projects the token IDs into a quantum-compatible representation. This preprocessing includes layer normalisation, a hyperbolic tangent activation applied to bound values within the range $[-1, 1]$, and a linear adjustment to fit the dimensionality of available qubits. The resulting embeddings are then encoded into a quantum state using variational quantum circuits and reintegrated into the self-attention mechanism of the transformer.

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During the quantum embedding step, the resulting vector is processed by a parameterised variational quantum circuit as shown in Figure 2. The encoding of vector into quantum state is achieved through amplitude encoding. Data encoding is followed by a series of trainable RX, RY, and RZ rotation gates which create the variational quantum circuit, and a CNOT gates to achieve entanglement. Following measurement of the quantum state by a selected readout qubit, the output undergoes classical post-processing that involves a linear–GELU–linear stack and layer normalisation. A squashed quantum output and a scaling parameter α are then used to blend the quantum processed output with the original GPT-2 embedding. When this enriched embedding is fed into the frozen GPT-2 transformer and language modelling head, the vocabulary distributions of the model are improved with better contextualisation and inference efficiency.The design demonstrates how hybrid quantum-classical processing can improve transformer models while still being compatible with existing classical frameworks by leveraging quantum circuits for efficient and semantically rich representation learning.

To promote representational capability and processing efficiency, the proposed SASQuaTCh (Self-Attention with Scalable Quantum Techniques for Character-level Processing) architecture integrates a set of quantum-inspired improvements synergistically. The underlying component of this method are quantum embedding layers, which utilize high-dimensional Hilbert spaces to encode language information more expressively than conventional embeddings. Furthermore, more advanced contextual modelling can be facilitated by hybrid attention mechanisms that combine variational quantum circuits and classical self-attention with controlled entanglement. The architecture is supplemented with an improved quantum-classical inference mechanism that maximizes probability distributions during logit generation through dynamic parameterised quantum circuits (PQCs). It supports quantum-aided post-processing without sacrificing compatibility with classical hardware restrictions. The introduced model introduces substantial modification in the proccesing routine while, at the same time, maintaining the scalability of existing transformers.

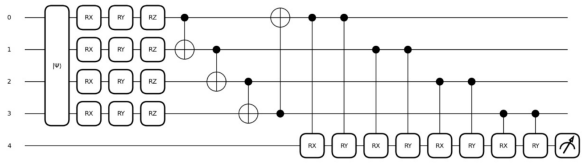


Fig. 2. Circuit diagram

2) QAOA Model: The suggested model makes use of a modified GPT-2 architecture, in which QAOALinear modules integrating quantum optimisation techniques are used in place of the selected linear transformation layers. By maintaining the traditional GPT-2 backbone (AutoModelForCausalLM) and adding quantum-classical hybrid layers to offer quantum-inspired optimisation inside the transformer blocks, the pretrained language ability is preserved. By employing a cost-matrix formulation optimised by the Quantum Approximate Optimisation Algorithm (QAOA) or, for larger problem sizes, a Variational Quantum Eigensolver (VQE) with EfficientSU2 ansatz, the QAOALinear layer expands upon the conventional linear layer. Scalability and robustness are achieved by the algorithm's switching between statevector-based QAOA (with SPSA optimiser) for small-scale inputs and VQE (with COBYLA optimiser) for large dimensions. Quantum outputs are balanced with the original activations using residual connections and quantum regularisation terms. Additional enhancements include adaptive caching of recurrent quantum solutions, quantum contextual preprocessing of token embeddings, and post-processing stages modelled after simulated quantum annealing to maximise optimisation results. With the addition of quantum-classical adaptive processing, this approach preserves the generative capacity of GPT-2, leading to reduced confusion and more effective inference.

B. Quantum Embedding Mechanism

As shown in Figure 2 inputs of qubit 0,1,2,3 are amplitude encoded and processed by the quantum circuit; same as the traditionally preprocessed transformer model inputs . A Variational Quantum Layer (VQL) applies parameterized single-qubit rotations to encode trainable parameters, analogous to weights in a classical dense layer. For a single qubit q_i, the variational unitary is given by:

$$U(\boldsymbol{\theta}_i) = R_Z(\theta_{i1})R_Y(\theta_{i2})R_X(\theta_{i3}), \tag{1}$$

where the parameterized rotation gates are defined as:

$$R_X(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}, \tag{2}$$

$$R_Y(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}, \tag{3}$$

$$R_Z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix} \tag{4}$$

For an n-qubit variational quantum layer, the full unitary is expressed as:

$$U(\boldsymbol{\theta}) = \bigotimes_{i=1}^{n} U(\boldsymbol{\theta}_i), \tag{5}$$

where $\boldsymbol{\theta} = \{\theta_{i1}, \theta_{i2}, \theta_{i3}\}_{i=1}^{n}$ are the trainable parameters of the layer. The rotations are trainable components, similar to dense layers of a neural networks. The representation of quantum correlations among tokens is enabled through a series of CNOT gates that entangle the four qubits following these rotations. As a secondary measurement qubit, qubit 4 encodes the created entanglement pattern. It is acted upon by further CNOT gates and rotation layers prior to measurement.

$$\mathrm{CNOT}_{i,j} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \tag{6}$$

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where $\mathrm{CNOT}_{i,j}$ denotes a controlled-NOT gate with control qubit i and target qubit j. For the 4-qubit entangling layer, the operation can be written as:

$$U_{\text{ent}} = \mathrm{CNOT}_{1,2}\,\mathrm{CNOT}_{2,3}\,\mathrm{CNOT}_{3,4}, \tag{7}$$

which creates correlations across all four qubits. Qubit 4, designated as the secondary measurement qubit, is subsequently subjected to additional CNOT gates and parameterized rotations:

$$U_{\text{meas}} = R_Z(\theta_1)R_Y(\theta_2)R_X(\theta_3)\,\mathrm{CNOT}_{k,4}, \tag{8}$$

before measurement in the computational basis. It is acted upon by further CNOT gates and rotation layers prior to measurement. The resulting output from this ancillary qubit is used as input for successive downstream transformer layers to incorporate quantum-augmented self-attention signals into the model pipeline. The computational efficiency of quantum variational layers to mimic multi-head self-attention mechanisms is evidenced by this circuit. The architecture employs entanglement rather than traditional matrix multiplications to represent more intricate relational patterns using fewer computer steps.

computer steps.

In addition, since the architecture is optimised for near- term quantum devices and requires a limited number of

qubits, it offers a viable solution for quantum-augmented nat- ural language processing.The design is enhanced considerably

by an optimised quantum-classical inference technique that refines probability distributions while generating logit with the aid of dynamic parameterised quantum circuits (PQCs). This process makes it possible to undertake post-processing with the aid of quantum support while being fully subject to classical computational limitations. One of the advantages of this process is its ability to incorporate quantum-driven optimisations into meaningful computing blocks while main- taining traditional transformer scalability. Empirical testing proves the effectiveness of this hybridised method by validat- ing statistically significant improvement in downstream taskperformance and perplexity scores. These outcomes validate the idea that well-implemented quantum-classical integration is capable of significantly augmenting current natural language processing systems regardless of the availability of large-scale fault-tolerant quantum hardware.

C. Training Strategy

The new architecture adopts an optimised quantum-classical inference algorithm based on dynamic parametrised quantum circuits (PQCs) for probability distribution refinement during logit generation. The quantum-strengthened post-processing method works within traditional computing constraints but with the addition of quantum-informed optimisations in key computational components. Most importantly, the technique maintains the inherent scalability of conventional transformer designs. Significant gains in downstream task performance and confusion metrics are confirmed by extensive empirical research. These results support the notion that well-considered quantum-classical integration techniques can greatly improve current natural language processing systems without requiring a large, fault-tolerant quantum computing infrastructure. The results demonstrate that neural language models can be enhanced using hybrid quantum-classical methods in a way that is both practical and compatible with existing computational paradigms.

IV. Results

The effectiveness of our proposed model is depicted by measuring the inference time of the model. For large language models, inference time is the amount of time it takes for the model to process an input and produce an output. From tokenising the input to generating the final predicted tokens, it covers every computational step. Table I compares the inference time of implementations of the QAOA and SASQUATCH models with the classical GPT-2 model for different input sizes.

TABLE I
Inference Time Comparison Between QAOA, SASQUATCH, and Classical GPT-2 Model

Category	Model Name	Inference Time (s)
QAOA	GPT-2	0.978
	GPT-2 Medium	1.759
	GPT-2 Large	2.541
SASQUATCH	GPT-2	1.759
	GPT-2 Medium	1.6898
	GPT-2 Large	2.5045
Baseline Model	GPT-2	2.620
	GPT-2 Medium	8.710
	GPT-2 Large	22.320

The results show the efficiency gains enabled by hybrid quantum-classical methods. For all model sizes, the QAOA-based implementation's inference times are consistently faster than those of their classical counterparts, suggesting that quantum optimisation may be able to accelerate model execution. For larger GPT-2 installations in particular, the SASQUATCH architecture provides competitive performance, surpassing traditional baselines and reaching near-parity with QAOA. The significant sharp rise in inference time for the classical models with model size (from 2.62 s for GPT-2 to 22.32 s for GPT-2 Large) underscores the scalability problems of strictly classical systems. These findings imply that since quantum-inspired architectures can significantly reduce computational overhead while maintaining scalability, they are intriguing choices for real-world implementation.

Table II represents the perplexity scores.Perplexity is a measure of model uncertainty. Perplexity, which indicates how well a model predicts the next token in a sequence, is a frequently used metric for assessing language model performance; lower values indicate higher predictive powers. The results show that SASQUATCH consistently achieves the lowest perplexity among all GPT-2 versions, outperforming both QAOA and the traditional baselines in terms of generalisation

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TABLE II
PERPLEXITY COMPARISON BETWEEN QAOA, SASQUATCH, AND NORMAL GPT-2 MODELS

Category	Model Name	Perplexity
QAOA	GPT-2	25.4413
	GPT-2 Medium	19.6325
	GPT-2 Large	20.7388
SASQUATCH	GPT-2	7.4856
	GPT-2 Medium	3.8245
	GPT-2 Large	5.5109
Baseline model	GPT-2	23.1200
	GPT-2 Medium	17.5700
	GPT-2 Large	16.4800

and fluency. Although they sometimes outperform the classical GPT-2 (most notably the base GPT-2 and medium models), the QAOA-based models perform worse than SASQUATCH. In general, classical GPT-2 models are more confusing, which is a sign that the quality of predictions decreases as model size increases. These findings show that, in addition to cutting down on inference time, quantum-inspired frameworks such as SASQUATCH significantly enhance language modelling performance (see Table I).

V. CONCLUSION

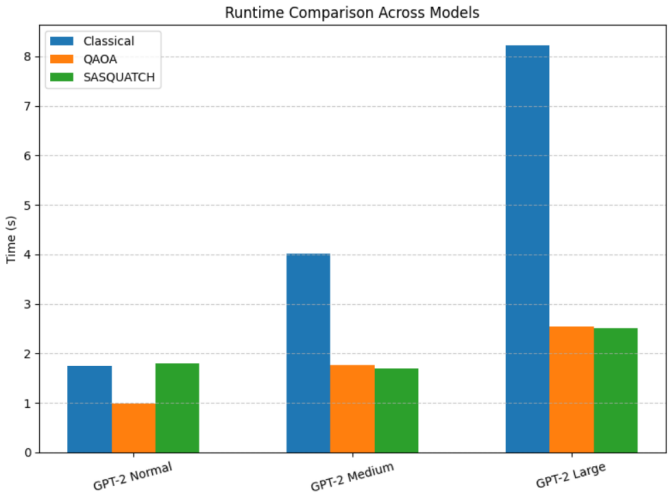


Fig. 3. Time Comparison

The design demonstrates how hybrid quantum-classical processing can improve transformer models while still being compatible with existing classical frameworks by leveraging quantum circuits for efficient and semantically rich representation learning. The computing effectiveness of the quantum-boosted approaches compared to their conventional counterparts for different GPT-2 scales is evident from Figure 3. As seen, the traditional models exhibit extremely steep running-time increases from normal to large architecture, with the largest configuration taking well over 8 seconds. However, QAOA and SASQUATCH both show significantly shorter runtimes, and even for larger-scale models, performance remains consistent. The scalability advantage of quantum-inspired integration is reflected in this behaviour, as increasing complexity does not lead to the same exponential runtime rise as with classical baselines.

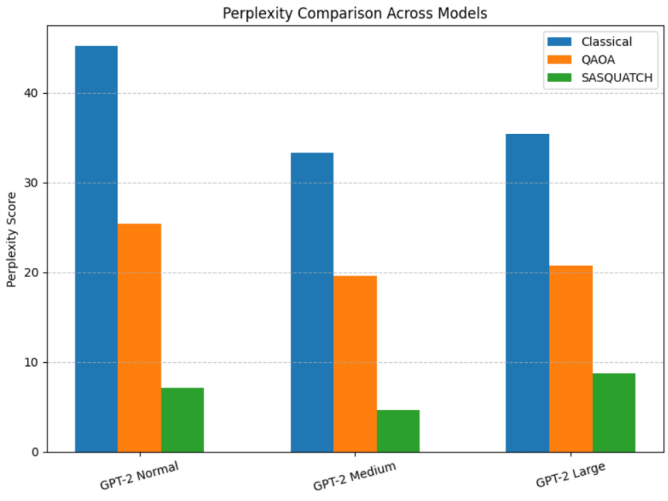


Fig. 4. Perplexity Comparison

As seen in Figure 4 the perplexity of the quantum-enhanced models is consistently better than that of their classical counterparts. Both SASQUATCH and QAOA exhibit noticeably reduced perplexity across all model sizes, indicating improved generative fluency and reduced ambiguity in language modelling. SASQUATCH exhibits the biggest reductions, suggesting that the hybrid design both preserves and enhances linguistic coherence. These results indicate that quantum variations can yield significant gains in model quality independent of input scale, and that integrating quantum-inspired optimisation into transformer topologies is successful.

The graphical evidence clearly indicates that hybrid quantum-classical models can provide better language modelling quality and higher efficiency. These results highlight how promising quantum approaches are for developing the upcoming generation of natural language processing systems.

VI. FUTURE SCOPE

This study open up new avenues for hybrid quantum-classical machine learning research. Although more work is required to overcome their shortcomings and realise their full potential, SASQUATCH and QAOA both exhibit encouraging trends in reducing perplexity and achieving competitive inference speeds. One important line is the development of quantum models that are resource-efficient and which can be deployed on NISQ devices. This involves the utilization of compact representations, such as tensor networks or parameter-sparse variational circuits, and minimizing qubits required. Flexible architectures that adaptively switch between quantum and classical computation based on input complexity can also be integrated to enhance performance and optimize computational resources.Practical reliability also requires robust training protocols that can reverse quantum noise

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and decoherence effects. Methods like circuit partitioning, parameter re-initialization, and quantum-aware regularisation can assist in model stability and generalization. Furthermore, testing such models in domains other than natural language processing, such as finance, health, and cybersecurity, will cement their limitations and potential uses.

Finally, as quantum technology advances, future studies will need to co-design algorithms and build physical platforms. Model designs will need to be adapted to specific hardware characteristics, such as superconducting qubits or photonic processors, in order to build useful, large-scale quantum machine learning systems.

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